

Lab 4: Orthogonality and Projections **Solutions**

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 23rd, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section. To receive credit for Activity 1, you'll need to show your lab TA that all test cases have passed **and** that you have answered the written questions in Task 4. Instructions on how to do this are in the lab notebook.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Projections

- ([Chapter 3.4](#)) The **orthogonal projection** of the vector $\vec{u}$ onto the vector $\vec{v}$ is given by

$$\vec{p} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}$$

Above, the scalar $k^* = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}$ was chosen to minimize $\|\vec{u} - k\vec{v}\|^2$.

- The vector $\vec{p}$ is called the orthogonal projection because the resulting error vector,

$$\vec{e} = \vec{u} - \vec{p} = \vec{u} - k^* \vec{v}$$

is orthogonal to $\vec{v}$, meaning $\vec{e} \cdot \vec{v} = 0$.

Activity 1: Presidential Speeches and Cosine Similarity

Complete the tasks in the `lab04.ipynb` notebook.

There are two ways to access the supplemental Jupyter Notebook:

- **Option 1 (preferred):** Set up a Jupyter Notebook environment locally, use `git` to clone our [course repository](#), and open `labs/lab04/lab04.ipynb`. For instructions on how to do this, see the [Environment Setup](#) page of the course website.
- **Option 2:** Click [here](#) to open `lab04.ipynb` on DataHub. Before doing so, read the instructions on the [Environment Setup](#) page on how to use the DataHub.

To receive credit for Activity 1, you'll need to show your lab TA that all test cases have passed **and** that you have answered the written questions in Task 4.

Activity 2: Sum-Difference Orthogonality

Let $\vec{u} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 5 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ -3 \end{bmatrix}$.

- a) Show that $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ are orthogonal.

Solution:

Let's start by computing the two vectors:

$$\vec{u} + \vec{v} = \begin{bmatrix} 3 \\ 1 \\ 4 \\ 2 \end{bmatrix}, \quad \vec{u} - \vec{v} = \begin{bmatrix} 1 \\ -3 \\ -4 \\ 8 \end{bmatrix}.$$

Their dot product is

$$(\vec{u} + \vec{v}) \cdot (\vec{u} - \vec{v}) = 3 \cdot 1 + 1 \cdot (-3) + 4 \cdot (-4) + 2 \cdot 8 = 3 - 3 - 16 + 16 = 0.$$

Since the dot product is 0, the vectors are orthogonal.

- b) Now suppose $\vec{u}, \vec{v} \in \mathbb{R}^n$ are arbitrary vectors with the same number of components. Is it always true that $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ are orthogonal?

- If so, prove why.
- If not, specify conditions under which it's guaranteed that $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ are orthogonal.

Hint: Use the distributive property of the dot product, which states that

$$(\vec{a} + \vec{b}) \cdot (\vec{c} + \vec{d}) = \vec{a} \cdot \vec{c} + \vec{a} \cdot \vec{d} + \vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{d}$$

Solution:

For any two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$,

$$(\vec{u} + \vec{v}) \cdot (\vec{u} - \vec{v}) = \vec{u} \cdot \vec{u} - \vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{u} - \vec{v} \cdot \vec{v} = \|\vec{u}\|^2 - \|\vec{v}\|^2,$$

since $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$.

So, in order for $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ to be orthogonal, we need

$$\|\vec{u}\|^2 - \|\vec{v}\|^2 = 0$$

which means

$$\|\vec{u}\| = \|\vec{v}\|$$

So, $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ are orthogonal if (and only if!) the two vectors have equal length. That was the case in part a) — both vectors had a norm of $\sqrt{2^2 + (-1)^2 + 0^2 + 5^2} = \sqrt{30}$.

Activity 3: Orthogonal Projections

Let $\vec{c} = \begin{bmatrix} 1 \\ 2 \\ -4 \\ 0 \end{bmatrix}$ and $\vec{d} = \begin{bmatrix} 3 \\ 2 \\ 0 \\ -1 \end{bmatrix}$.

- a) Find the orthogonal projection of $\vec{c}$ onto $\vec{d}$. Call this vector $\vec{q}$.

Solution:

$$\begin{aligned}\vec{q} &= \left(\frac{\vec{c} \cdot \vec{d}}{\vec{d} \cdot \vec{d}} \right) \vec{d} \\ &= \frac{1 \cdot 3 + 2 \cdot 2 + (-4) \cdot 0 + 0 \cdot (-1)}{3^2 + 2^2 + 0^2 + (-1)^2} \vec{d} \\ &= \frac{3 + 4}{9 + 4 + 1} \vec{d} \\ &= \frac{7}{14} \vec{d} \\ &= \frac{1}{2} \vec{d} \\ &= \begin{bmatrix} 1.5 \\ 1 \\ 0 \\ -0.5 \end{bmatrix}\end{aligned}$$

- b) Find the error vector, $\vec{r} = \vec{c} - \vec{q}$. Which vector is $\vec{r}$ orthogonal to, $\vec{c}$ or $\vec{d}$? Draw a rough picture of the relationship between $\vec{c}$, $\vec{d}$, $\vec{q}$, and $\vec{r}$. You may want to review [Chapter 3.4](#).

Solution: First, let's find $\vec{r}$.

$$\begin{aligned}\vec{r} &= \vec{c} - \vec{q} \\ &= \begin{bmatrix} 1 \\ 2 \\ -4 \\ 0 \end{bmatrix} - \begin{bmatrix} 1.5 \\ 1 \\ 0 \\ -0.5 \end{bmatrix} \\ &= \begin{bmatrix} -0.5 \\ 1 \\ -4 \\ 0.5 \end{bmatrix}\end{aligned}$$

$\vec{r}$ is orthogonal to $\vec{d}$, not $\vec{c}$, as confirmed by the dot products. The key idea we introduced in [Chapter 3.4](#) is that the error vector is orthogonal to the vector we projected onto. Here, $\vec{r}$ is the error vector and $\vec{d}$ is the vector we projected onto.

$$\begin{aligned}\vec{r} \cdot \vec{c} &= (-0.5) \cdot 1 + 1 \cdot 2 + (-4) \cdot (-4) + 0.5 \cdot 0 \\ &= -0.5 + 2 + 16 \\ &= 17.5\end{aligned}$$

$$\begin{aligned}\vec{r} \cdot \vec{d} &= (-0.5) \cdot 3 + 1 \cdot 2 + (-4) \cdot 0 + 0.5 \cdot (-1) \\ &= -1.5 + 2 - 0.5 \\ &= 0\end{aligned}$$

Activity 4: Orthogonal Decomposition with Orthonormal Vectors

Let

$$\vec{v}_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$\vec{v}_1$ and $\vec{v}_2$ are called **orthonormal**, because they are:

- unit vectors: $\|\vec{v}_1\| = 1$ and $\|\vec{v}_2\| = 1$.
- orthogonal: $\vec{v}_1 \cdot \vec{v}_2 = 0$.

Additionally, let $\vec{u} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$.

Our goal in this activity is to write $\vec{u}$ as a linear combination of $\vec{v}_1$ and $\vec{v}_2$. The fact that $\vec{v}_1$ and $\vec{v}_2$ are orthonormal makes this simple.

a) Find $\vec{u} \cdot \vec{v}_1$ and $\vec{u} \cdot \vec{v}_2$. Your answer should involve $\sqrt{5}$; don't use a calculator.

Solution: To find each dot product, we multiply corresponding components and add:

$$\vec{u} \cdot \vec{v}_1 = 4 \cdot \frac{1}{\sqrt{5}} + (-1) \cdot \frac{2}{\sqrt{5}} = \frac{4-2}{\sqrt{5}} = \boxed{\frac{2}{\sqrt{5}}}$$

$$\vec{u} \cdot \vec{v}_2 = 4 \cdot \left(-\frac{2}{\sqrt{5}}\right) + (-1) \cdot \frac{1}{\sqrt{5}} = \frac{-8-1}{\sqrt{5}} = \boxed{-\frac{9}{\sqrt{5}}}$$

b) Evaluate

$$(\vec{u} \cdot \vec{v}_1)\vec{v}_1 + (\vec{u} \cdot \vec{v}_2)\vec{v}_2$$

What do you notice?

Solution: Using the dot products we found in part a), we have

$$\begin{aligned} (\vec{u} \cdot \vec{v}_1)\vec{v}_1 + (\vec{u} \cdot \vec{v}_2)\vec{v}_2 &= \frac{2}{\sqrt{5}} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} - \frac{9}{\sqrt{5}} \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{5} \\ \frac{4}{5} \\ \frac{2}{5} \end{bmatrix} + \begin{bmatrix} \frac{18}{5} \\ \frac{9}{5} \\ \frac{9}{5} \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ -1 \end{bmatrix} = \boxed{\vec{u}} \end{aligned}$$

We get back $\vec{u}$! In other words, the dot products from part a) are exactly the coefficients we need to write $\vec{u}$ as a linear combination of $\vec{v}_1$ and $\vec{v}_2$.

c) Why is $(\vec{u} \cdot \vec{v}_1)\vec{v}_1 + (\vec{u} \cdot \vec{v}_2)\vec{v}_2 = \vec{u}$?

Hint: Start by drawing a picture of $\vec{u}$, $\vec{v}_1$, and $\vec{v}_2$, and reviewing the “Orthogonal Decomposition” section of Chapter 3.4 and the very last example discussed in yesterday’s lecture.

Solution: The two terms in part b) are the projections of $\vec{u}$ onto the orthogonal directions $\vec{v}_1$ and $\vec{v}_2$. Since these vectors are orthogonal, their projections add to $\vec{u}$, just as in the orthogonal decomposition we saw in lecture.

Activity 5: Orthogonal Decomposition with Non-Unit Vectors

- a) Let $\vec{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$ and $\vec{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$. Write $\vec{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ as a linear combination of $\vec{v}_1$, $\vec{v}_2$, and $\vec{v}_3$, and verify that your answer is correct. Note that $\vec{v}_1$, $\vec{v}_2$, and $\vec{v}_3$ are pairwise orthogonal.

Solution:

We are given

$$\vec{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} \quad \vec{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

We're looking for scalars a, b, c such that $\vec{u} = a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3$.

Solution 1: Solving a system of equations

$$a \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} + c \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Is equivalent to the system of equations:

$$-a + 2b + 2c = 1 \tag{1}$$

$$2a + 2b - c = 1 \tag{2}$$

$$2a - b + 2c = 1 \tag{3}$$

Using elimination:

$$\begin{aligned} (\text{Eq. 2}) - (\text{Eq. 3}) : \quad & (2a - 2a) + (2b - (-b)) + (-c - 2c) = 0 \\ \Rightarrow \quad & 3b - 3c = 0 \quad \Rightarrow \quad b = c \end{aligned}$$

$$\begin{aligned} (\text{Eq. 2}) - (\text{Eq. 1}) : \quad & (2a - (-a)) + (2b - 2b) + (-c - 2c) = 0 \\ \Rightarrow \quad & 3a - 3c = 0 \quad \Rightarrow \quad a = c \end{aligned}$$

This tells us that $a = b = c$. Plugging this back into Eq. 2 gives us:

$$-a + 2a + 2a = 1 \rightarrow 3a = 1 \rightarrow a = \frac{1}{3}$$

So, $a = b = c = \frac{1}{3}$, and:

$$\vec{u} = \frac{1}{3} \vec{v}_1 + \frac{1}{3} \vec{v}_2 + \frac{1}{3} \vec{v}_3$$

We can verify that we did this correctly by computing the right-hand side above:

$$\frac{1}{3} \vec{v}_1 + \frac{1}{3} \vec{v}_2 + \frac{1}{3} \vec{v}_3 = \frac{1}{3} \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1/3 + 2/3 + 2/3 \\ 2/3 + 2/3 - 1/3 \\ 2/3 - 1/3 + 2/3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \vec{u}$$

Solution 2: Using the fact that $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are orthogonal

As is alluded to in part b), we can use the fact that $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are orthogonal to find coefficients a, b , and c by projecting $\vec{u}$ onto each of the $\vec{v}_i$ s. This is similar to what was done in the [Orthogonal Decomposition](#) section of Chapter 3.4.

Let $\vec{p}_i$ be the projection of $\vec{u}$ onto $\vec{v}_i$, for $i = 1, 2, 3$. Then, we have:

$$\vec{p}_1 = \frac{\vec{u} \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 = \frac{1 \cdot (-1) + 1 \cdot 2 + 1 \cdot 2}{(-1)^2 + 2^2 + 2^2} \vec{v}_1 = \frac{3}{9} \vec{v}_1 = \frac{1}{3} \vec{v}_1$$

$$\vec{p}_2 = \frac{\vec{u} \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2 = \frac{1 \cdot 2 + 1 \cdot 2 + 1 \cdot (-1)}{2^2 + 2^2 + (-1)^2} \vec{v}_2 = \frac{3}{9} \vec{v}_2 = \frac{1}{3} \vec{v}_2$$

$$\vec{p}_3 = \frac{\vec{u} \cdot \vec{v}_3}{\vec{v}_3 \cdot \vec{v}_3} \vec{v}_3 = \frac{1 \cdot 2 + 1 \cdot (-1) + 1 \cdot 2}{2^2 + (-1)^2 + 2^2} \vec{v}_3 = \frac{3}{9} \vec{v}_3 = \frac{1}{3} \vec{v}_3$$

Adding $\vec{p}_1, \vec{p}_2, \vec{p}_3$ gives us:

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = \frac{1}{3} \vec{v}_1 + \frac{1}{3} \vec{v}_2 + \frac{1}{3} \vec{v}_3 = \vec{u}$$

- b) In general, suppose that $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ are **orthogonal** vectors in $\mathbb{R}^n$, meaning that $\vec{v}_i \cdot \vec{v}_j = 0$ for all $i \neq j$. **Given that** it is possible to write $\vec{u}$ as a linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$, show that the coefficients of the linear combination

$$\vec{u} = a_1 \vec{v}_1 + a_2 \vec{v}_2 + \dots + a_d \vec{v}_d$$

are given by

$$a_i = \frac{\vec{u} \cdot \vec{v}_i}{\vec{v}_i \cdot \vec{v}_i}$$

Hint: Start by taking the dot product of both sides of the linear combination equation with $\vec{v}_1$. What do you notice?

Solution:

We're told to assume that any pair of vectors among $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ are orthogonal, and that $\vec{u}$ can be written as a linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$.

$$\vec{u} = a_1\vec{v}_1 + a_2\vec{v}_2 + \dots + a_d\vec{v}_d$$

As the hint suggests, let's take the dot product of both sides with $\vec{v}_i$, where i is some value in $\{1, 2, \dots, d\}$.

$$\vec{u} \cdot \vec{v}_i = (a_1\vec{v}_1 + \dots + a_d\vec{v}_d) \cdot \vec{v}_i$$

Since $\vec{v}_i \cdot \vec{v}_j = 0$ for $i \neq j$, only the $i = j$ term survives:

$$\begin{aligned} \vec{u} \cdot \vec{v}_i &= (a_1\vec{v}_1 + \dots + a_d\vec{v}_d) \cdot \vec{v}_i \\ &= a_1(\vec{v}_1 \cdot \vec{v}_i) + \dots + a_{i-1}(\vec{v}_{i-1} \cdot \vec{v}_i) + a_i(\vec{v}_i \cdot \vec{v}_i) + a_{i+1}(\vec{v}_{i+1} \cdot \vec{v}_i) + \dots + a_d(\vec{v}_d \cdot \vec{v}_i) \\ &= a_1(0) + \dots + a_{i-1}(0) + a_i(\vec{v}_i \cdot \vec{v}_i) + a_{i+1}(0) + \dots + a_d(0) \\ &= a_i(\vec{v}_i \cdot \vec{v}_i) \end{aligned}$$

Solving for a_i above gives us

$$\vec{u} \cdot \vec{v}_i = a_i(\vec{v}_i \cdot \vec{v}_i) \implies a_i = \frac{\vec{u} \cdot \vec{v}_i}{\vec{v}_i \cdot \vec{v}_i}$$

Since i was arbitrary, the same calculation holds for any value of i in $\{1, 2, \dots, d\}$.

What we proved here in part **b)** is that when writing a vector $\vec{u}$ as a linear combination of orthogonal vectors, the coefficients of the linear combination can be found by projecting the vector $\vec{u}$ onto each of the orthogonal vectors and adding the results, rather than solving a system of equations.

c) In Activity 4, why didn't we divide by $\vec{v}_1 \cdot \vec{v}_1$ (and $\vec{v}_2 \cdot \vec{v}_2$) when finding the coefficients?

Solution: The vectors in Activity 4 are unit vectors, so $\vec{v}_1 \cdot \vec{v}_1 = \|\vec{v}_1\|^2 = 1$, and likewise for $\vec{v}_2$. Dividing by these dot products would just mean dividing by 1.

Activity 6: A Plane from Spanning Vectors

This activity previews ideas that we will explore in upcoming lectures.

An important idea from [Chapter 4.1](#) is that two non-parallel vectors in $\mathbb{R}^n$ (where $n \geq 2$) span a plane in n -dimensional space. Here, we'll show you how to find the equation of such a plane, given two vectors in $\mathbb{R}^3$. This is also touched on in [Chapter 4.4](#).

Let

$$\vec{u} = \begin{bmatrix} 5 \\ -7 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}, \quad P = \text{span}(\{\vec{u}, \vec{v}\})$$

- a) Find a nonzero vector $\vec{n}$ that is orthogonal to both $\vec{u}$ and $\vec{v}$. Verify your answer using dot products.

Solution: Write $\vec{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Orthogonality requires

$$5a - 7b + 3c = 0, \quad 4a + b - 2c = 0.$$

The second equation gives $b = 2c - 4a$. Substituting into the first gives $33a - 11c = 0$, so $c = 3a$ and $b = 2a$.

Note that there are infinitely many choices of a , meaning there are infinitely many $\vec{n}$ that satisfy the constraints of the question. All we were asked for is **a** nonzero vector that is orthogonal to both $\vec{u}$ and $\vec{v}$; once we find one such vector, any scalar multiple of it (of which there are infinitely many!) will also be orthogonal to both $\vec{u}$ and $\vec{v}$. Since we just need one vector, let's keep it simple: pick $a = 1$. Then,

$$\vec{n} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Indeed, $\vec{n} \cdot \vec{u} = 5 - 14 + 9 = 0$ and $\vec{n} \cdot \vec{v} = 4 + 2 - 6 = 0$.

b) Suppose your vector is $\vec{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Verify that both $\vec{u}$ and $\vec{v}$ satisfy

$$ax + by + cz = 0$$

Then, graph your equation on Desmos, [desmos.com/3d](https://www.desmos.com/3d).

Solution: Using $\vec{n} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ from part a), the equation is

$$x + 2y + 3z = 0.$$

Substituting the components of $\vec{u}$ gives $5 + 2(-7) + 3(3) = 0$, and substituting the components of $\vec{v}$ gives $4 + 2(1) + 3(-2) = 0$. Thus, both vectors satisfy the equation. The graph in Desmos is the plane through the origin spanned by $\vec{u}$ and $\vec{v}$.